

SIH26100 - Frozen Blueprint

AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement | MoPNG / CPCL | Frozen
28 Sep 2026 | Idea submission deadline 30 Sep 2026

AI reads and advises. Fixed rules verify. The officer decides.

Contents. Section A: PPT content (non-technical, goes on the slides). Section B: Demo and tech stack (used to build, demo and answer questions; not for slides).

Frozen architecture

Tender + bidder documents + (simulated) portal data
-> AI extraction / matching (never authoritative)
-> structured facts
-> deterministic tender rule engine
-> PASS / FAIL / REVIEW / UNVERIFIABLE per requirement
-> Compliance Score + independent Risk Level
-> AI advisory (explains rule results; cannot change them)
-> Procurement Officer decides (justification required)
-> Audit record (hash-chained, tender + rule-set versions)

- Tender-specific rule engine, configured from the tender's stated requirements and maintained as a reviewed rule library.
- AI extraction and advisory only; no LLM in the decision path.
- Evidence-linked verification: every result shows requirement, evidence, extracted value, validation, reason.
- Simulated portal adapters, clearly labelled SIMULATED everywhere.
- Officer-controlled final decision. A requirement FAIL is not a disqualification.

Frozen rules

Item	Definition
Score value per result	PASS 1.0 REVIEW 0.5 UNVERIFIABLE 0.0 FAIL 0.0
Weights	Mandatory = 3, Optional = 1 (set per requirement in the tender config)
Score formula	$100 \times \text{sum}(\text{weight} \times \text{value}) / \text{sum}(\text{weight})$, over requirements that apply to the tender. UNVERIFIABLE stays in the denominator at 0, so a portal outage never inflates the score.
Risk: HIGH	Any mandatory requirement is FAIL
Risk: MEDIUM	Any mandatory REVIEW or UNVERIFIABLE, or any optional FAIL
Risk: LOW	Otherwise
Independence	Risk is computed from the states, not from the score. A banner shows "mandatory requirement failed" regardless of the number.
Ranking	Bidders are never ranked by compliance score. Price and procurement evaluation stay separate (L1 is decided by price).

Worked example. 3 mandatory (PASS, FAIL, REVIEW) + 2 optional (PASS, UNVERIFIABLE). Weights 3+3+3+1+1 = 11. Values: 3 + 0 + 1.5 + 1 + 0 = 5.5. Score = 50/100. Risk = HIGH (mandatory FAIL). Second case: all 3 mandatory PASS, optional PASS + UNVERIFIABLE: (9 + 1 + 0) / 11 = 91/100, Risk = LOW.

Frozen tiers

Tier	Items	Rule
A: must actually work	1 Rule engine. 2 GST/PAN/CIN/Udyam validators. 3 GSTIN-PAN cross-check. 4 GST registration + return-filing mock. 5 Four result states. 6 Score + risk rules. 7 Evidence viewer. 8 Audit hash chain. 9 Tender / rule-set versioning. 10 Advisory template + optional model.	Build first. Make this vertical slice extremely convincing.
B: only after A	Entity matching. DigiLocker mock. OEM extraction.	Start only when every Tier A item works end to end.

Tier	Items	Rule
C: do not let it consume build time	Deep ITR verification. Anomaly signals. Load testing. Additional portal depth.	Slide-only as "designed" unless time remains. Never show an unbuilt feature as working.

Change policy: if a new idea appears, the default answer is no, unless it directly improves a Tier A behaviour.

Sequence: build, measure, demo, PPT, rehearse judge questions.

SECTION A - PPT CONTENT

Plain language. No framework, database or model names on slides. Each slide answers one judge question at a glance.

Slide-by-slide (8 slides)

#	Slide	Content
1	Title	Project name, PS SIH26100, team. Promise line: "Every bidder claim checked against evidence; every decision traceable."
2	The problem	For one bid, an officer manually checks many bidder documents against about 8 separate government sources and the tender's requirements. Slow, inconsistent, error-prone. Visual: 1 bid -> documents -> government sources -> tender requirements -> manual cross-check.
3	The gap	Bid-specific evidence verification. GeM runs registration and bid mechanics. For each tender, the officer must still establish that the bidder's evidence is complete, consistent and compliant for that particular tender, across documents and sources. That cross-check is the gap. (Do not say "GeM does not verify anything".)
4	Our approach	Flow: Tender requirements -> Bidder documents + portal data -> AI extracts and finds inconsistencies -> Fixed tender rules verify -> Score + risk + advisory -> Officer decides. Footer line: AI reads and advises. Fixed rules verify. The officer decides. Add: "Rules are configured from the tender's stated requirements and maintained as a reviewed rule library."
5	What the officer sees	Dashboard mock: bidder name, Compliance Score (example 50/100), Risk Level (HIGH), counts of PASS / FAIL / REVIEW / UNVERIFIABLE, then requirement rows. One expanded row: GSTIN differs between two documents, both values side by side, [View evidence]. Caption: System verifies requirements. The officer makes the procurement decision.
6	Why it is trustworthy	(1) AI cannot override verification rules. (2) Missing or unavailable information is never counted as a pass. (3) Every check and decision is recorded, tamper-evident, with tender and rule-set versions. (4) Designed for on-premises deployment; documents need not leave the organisation.
7	Measured impact	Two separate boxes. PS TARGET: 60-80% reduction in verification effort. OUR PROTOTYPE: manual X min, system Y min, observed Z% (N bidder packs; manual baseline timed by teammates on the same packs). Also: false-PASS observed 0 of N; REVIEW rate X%; categories covered X of about 10. Bottom line: prototype uses synthetic data and simulated portal responses.
8	Roadmap and close	Pilot on one CPSE tender -> integrate through authorised channels -> extend the reviewed rule library. Rule sets are configured per organisation; the engine is shared, which gives standardised verification across CPSEs. Strip: <i>Built in prototype</i> vs <i>Designed, not built</i> . Close: "We don't ask an AI whether a bidder is compliant; we build a traceable chain from each tender requirement to its evidence, its validation, and the officer's decision."

Words to use vs avoid

Use	Avoid
Verification, evidence, traceable, officer-controlled, requirement-by-requirement	Revolutionary, intelligent, smart, predictive
AI reads and advises; fixed rules verify	AI decides / AI-powered compliance verdict
Tamper-evident record	Blockchain
Zero false-PASS observed on N synthetic cases	Zero errors / 100% accurate / zero false positives
Simulated portal responses (labelled)	Connected to GSTN / integrated with GeM
Designed for on-premises deployment	Our prototype runs on-premises
Specific inconsistency checks	Detects forged documents

Numbers to measure and place on slides

- False-PASS observed and number of cases tested.
- REVIEW rate (share of requirements or bidders routed to the officer).

- Manual vs system time per bidder pack, N, and how the baseline was obtained.
- Requirement categories covered out of about 10 in the PS.

Keep off slides: stack, database, queue and model names, thresholds, hash details. They are used to answer questions (Section B).

One-line answers for Q and A

Question	Answer
Are you connected to GSTN?	No. The PS expects portal integration, but live access is not available to the team and the PS permits dummy data. Simulated adapters are labelled; an authorised production adapter replaces them without touching the rule engine.
Where is the AI?	It reads messy documents, finds missing and inconsistent information, and writes the advisory. It never sets a result.
Why not let an LLM decide?	Decisions must be reproducible: the same evidence gives the same result.
Who decided the rules?	The tender's stated requirements, applied through a reviewed, versioned rule library.
Does FAIL mean disqualified?	No. It is a requirement result. Only the officer decides, with a written justification.
REVIEW vs UNVERIFIABLE?	REVIEW: evidence is incomplete or ambiguous. UNVERIFIABLE: the external check could not run.
Can you detect forged documents?	We detect defined inconsistencies and validation failures. We do not claim to identify every forgery.

SECTION B - DEMO AND TECH STACK

Everything technical. For building the prototype, running the demo, and answering judges. Not for slides.

1. PS coverage (14 expected-solution items)

#	PS item	Implementation	Tier
1	Portal integration	Adapter interface + SIMULATED implementations	A
2	Udyam / MSME	Format validator, document, mock lookup, tender applicability	A
3	GST registration and return filing	GSTIN validator; mock returns registration + filing status, last return, period, date	A
4	PAN + Income Tax	PAN validator + GSTIN-PAN cross-check; ITR acknowledgement extraction only if time	A / C
5	Make in India	Per-tender local-content threshold rule	A
6	EPFO / ESIC	Conditional rules by tender category; mock lookup	A
7	Startup / NSIC / OEM	Registration mocks; OEM letter extraction	A / B
8	DigiLocker	Simulated adapter: compare hash/fields of uploaded doc with issued record	B
9	Blacklisting / debarment	Mock debarment list with scope and dates	A
10	Tender-specific compliance	Versioned tender rule config	A
11	AI missing / inconsistent info	Extraction + missing-field detection (A); entity matching (B); anomaly signals (C)	A / B / C
12	Compliance Score + Risk	Frozen formulas above	A
13	AI recommendation	Deterministic advisory template; optional model rewrites wording	A
14	Audit record	Hash chain + tender / rule-set versions + evidence hashes	A

2. Why simulated portals (say exactly this)

"The prototype uses simulated portal adapters and dummy datasets because production portal access is not available to the hackathon team. Every simulated response is labelled."

- The PS says the platform shall integrate with portals, and separately states that dummy bidder and tender datasets may be used. Do not claim the PS forbids live integration.
- As far as we know, GSTN data reaches applications through licensed intermediaries under formal agreements, and taxpayer-level data needs authentication and consent. Not obtainable before 30 Sep.
- Do not scrape public portals (captcha, terms, fragile, integrity risk). Do not display "Connected to GSTN".
- Portal panel must read: SIMULATED ADAPTER, response, latency, fetched_at.
- Why the adapter boundary exists: external verification is a separate dependency, so we can demonstrate available, unavailable, delayed and stale evidence, and swap in a production adapter later.

```
class PortalAdapter:
    def lookup(self, id_value) -> PortalResult

PortalResult = {status, fields, fetched_at, latency_ms,
                source: 'SIMULATED' | 'LIVE'}

MockGSTN, MockUdyam, MockEPFO, MockDigiLocker ... # injectable
                                                    # delay / failure
```

GST mock response example: registration ACTIVE; return filing COMPLIANT | OVERDUE; last return GSTR-3B; period Aug 2026; filed 20 Sep 2026. Failure case: registration active but return filing overdue.

3. Pipeline and decision trace

```
PDF/scan -> sanitise -> extract fields -> validators
-> rule engine (tender config) -> result state
```

-> score + risk -> advisory -> officer decision -> audit log

REQ-07 Valid GST registration and return filing (mandatory)

Evidence: GST_Certificate.pdf (sha256 ...)

Extracted: GSTIN = 23ABCDE1234F1Z5 (page 1, region shown)

Validation: format OK | checksum OK | PAN match OK

portal: SIMULATED, ACTIVE, filing OVERDUE

Result: FAIL (mandatory)

Officer: [Approve] [Reject] [Request clarification]

4. Tender-specific rule configuration

```
tender: {id: CPCL-T-001, version: 2, rule_set_version: 1.3}
```

```
requirements:
```

```
- id: GST_001
```

```
  mandatory: true
```

```
  applies_if: tender.gst_required
```

```
  evidence: [GST_CERTIFICATE]
```

```
  checks: [GSTIN_CHECKSUM, GSTIN_PAN_MATCH,  
           PORTAL_ACTIVE, RETURN_FILING_CURRENT]
```

```
  on_fail: FAIL
```

```
  on_missing: REVIEW
```

```
  on_portal_down: UNVERIFIABLE
```

```
- id: MSME_001
```

```
  mandatory: false          # preference only in this tender
```

```
  applies_if: tender.msme_preference
```

```
  on_missing: PASS_NO_PREFERENCE
```

- Compile before coding: Udyam mandatory vs preference; Make in India local-content thresholds; EPFO/ESIC applicability by category; debarment scope (organisation-wide, category, time-bound); OEM authorisation fields (product, dealer, validity, signatory).
- Certificate validity is compared with the bid opening date, not today.
- Rule library is versioned and reviewed. We claim engine correctness and coverage, not legal completeness.
- Unit-test every rule with positive, negative and edge cases.

5. Result states and failure semantics

Situation	Result	Reason
All checks satisfied	PASS	Evidence attached.
Explicit violation (bad checksum, expired, mismatch, listed as debarred, filing overdue)	FAIL	Deterministic, explainable. Not a disqualification.
Required field missing, unreadable or ambiguous	REVIEW	Evidence incomplete. Missing evidence is never a pass.
Portal timeout / down	UNVERIFIABLE	External check could not run; retry with backoff; show cached snapshot with age.
LLM-extracted value not found verbatim in source text	REVIEW	Grounding rule: prevents hallucinated fields and OCR slips (5 vs S).

UI copy: **REVIEW** = we have the evidence but it is incomplete or ambiguous. **UNVERIFIABLE** = the required external verification could not be performed.

6. AI layer (bounded)

Component	What it does	Boundary
Extraction (A)	Regex/template first; OCR (Tesseract + OpenCV) for scans; model fallback for unstructured text such as the OEM letter.	Output is data. Value must appear verbatim in source text, else REVIEW.
Missing-info detection (A)	Compares required evidence per tender rule with what was extracted.	Emits findings; the rule engine sets the state.
Entity matching (B)	Normalise (casefold; strip M/s, Pvt, Private, Ltd) then token similarity (rapidfuzz).	Thresholds live in config, start values tuned on synthetic cases. Band between match and mismatch routes to REVIEW.

Component	What it does	Boundary
Anomaly signals (C)	Metadata, hidden text, font inconsistencies.	Feeds REVIEW flags only.
Advisory (A)	A deterministic template builds the officer summary from rule results. An optional model rewrites the wording.	Input is structured results only. Output must cite existing requirement IDs; if validation fails, fall back to the template. Labelled advisory.

- **Model for the demo:** either a small local model (Ollama) on the recording laptop, or a hosted API on synthetic data only, stated plainly; production is designed for a local model. The template guarantees the advisory always works.
- **Injection:** no LLM in the decision path, and the advisory sees only structured results, so hidden instructions in a PDF cannot change a state.
- **Deterministic cross-check:** characters 3 to 12 of a GSTIN are the holder's PAN; compare with the PAN document.
- **Synthetic data:** generate GSTINs with valid checksums, or your own validator will fail your own dataset.

7. Audit trail

```
record_n = { tender_id, tender_version, rule_set_version,
             requirement_results, evidence_hashes (sha256),
             score, risk, advisory_text_hash,
             officer_decision + justification, timestamp,
             prev_hash, hash = sha256(prev_hash + payload) }
```

- Database role for audit table: INSERT only (no UPDATE/DELETE). Verify endpoint recomputes the chain and reports the first broken record with expected vs computed hash.
- A document hash proves the exact file used is identifiable later; it does not prove the original document is genuine.
- Reason for a hash chain over a plain audit table: the table records what happened; the chain lets us detect that history was modified afterwards.

8. Tech stack (frozen)

Layer	Choice	Note
API	FastAPI (Python)	Background tasks for extraction jobs; Redis queue only if time remains.
Extraction	PyMuPDF / pypdf; Tesseract + OpenCV; rapidfuzz	Regex/template first; model fallback.
Rule engine	Python module, YAML rules	Pure functions, unit-tested, no LLM.
Database	PostgreSQL only	Transactional + audit (insert-only role).
Frontend	React / Next.js	Score, risk, state counts, requirement table, evidence viewer with highlighted region, officer workflow, RBAC-aware.
Roles	officer, evaluator, admin, auditor (read-only)	RBAC on API and UI.
Security	File type/size validation; text sanitisation; encryption at rest; retention policy	Aligned with DPDP Act 2023 principles.
Deploy	docker compose; hosted demo (Render / Vercel / Railway)	Hosted demo is not on-prem; state "designed for on-premises deployment".

9. Synthetic dataset and evaluation

- 3 tenders with distinct rule combinations (goods; services needing EPFO/ESIC; MSME-preference tender).
- 20+ bidder packs built as variants of each case: compliant; missing document; expired certificate; GSTIN-PAN mismatch; GST filing overdue; debarred; ambiguous (REVIEW); portal down (UNVERIFIABLE).
- Documents rendered as realistic PDFs/scans (noise, skew, stamps) so OCR is genuinely exercised.

- Report deterministic validation results per check separately from OCR extraction accuracy per field. Do not merge into one "AI accuracy".
- Wording: "zero false-PASS observed across N synthetic cases". No percentages from tiny samples.
- Manual baseline: teammates time manual verification of the same packs; state N and method.

Check	Tested	Correct	False	PASS	REVIEW
GSTIN validity	24	24	0	0	0
GSTIN-PAN match	20	20	0	0	0
Return filing	12	12	0	0	0
Identity match (B)	30	28	0	0	2

10. Demo script (3 min, Tier A only)

Time	What happens on screen
0:00-0:20	Problem: one bid, many documents, many sources, manual cross-check.
0:20-1:10	Bidder A: PASS, score high, risk LOW. Bidder B: GSTIN-PAN mismatch, FAIL on a mandatory requirement, risk HIGH; click through to both values side by side. Bidder C: missing certificate, REVIEW.
1:10-1:30	GST registration active but return filing overdue: FAIL with the filing evidence from the SIMULATED adapter.
1:30-1:50	Kill the simulated portal: result becomes UNVERIFIABLE, not PASS; cached snapshot and age shown; score drops accordingly.
1:50-2:15	Advisory panel: summary citing REQ IDs, labelled advisory. Officer chooses Request clarification; justification is mandatory.
2:15-2:40	Edit an old audit row in the database; click Verify Audit: chain broken at record N with expected vs computed hash. Show tender and rule-set version in the record.
2:40-3:00	Numbers on screen: false-PASS observed, REVIEW rate, time per bidder vs manual baseline. Close with the one-line identity statement.

Optional if time: poisoned PDF with hidden instruction text, showing the decision unchanged. Tier B items appear in the demo only if they fully work.

11. Build plan (2 days)

Block	Deliverable
Day 1 AM	Schema; rule engine; tender/rule-set versioning; validators (GSTIN, PAN, CIN, Udyam); GSTIN-PAN check; unit tests.
Day 1 PM	Mock adapters (GST registration + filing, Udyam, EPFO/ESIC, debarment) with injectable failure; extraction on synthetic PDFs; four states; score + risk.
Day 2 AM	Dashboard: score, risk, counts, requirement table, evidence viewer, officer workflow; audit hash chain + verify; advisory template.
Day 2 PM	Evaluation table and baseline timing; optional model for advisory; hosted deploy; limitations page; record video; finish PPT.
Tier B / C	Only after every Tier A item works end to end. Otherwise mark "designed" on slides.

If time is short, protect in this order: rule engine, validators, four states, score + risk, evidence view, audit chain, evaluation numbers, video. The PPT and video decide selection; the prototype supports them.

12. Hostile-judge answers (technical)

Question	Answer
What gap exists beyond GeM?	Bid-specific cross-verification of evidence across documents, sources and tender requirements.
Where do rules come from?	Tender's stated requirements; reviewed, versioned rule library; per-tender applicability is configured, not generated.
OCR misreads a GSTIN. Then?	Checksum and GSTIN-PAN cross-check fail or the value is not grounded in source text; result is FAIL or REVIEW, never PASS.
Why not an LLM decision?	Reproducibility: same evidence, same result. LLM is not in the decision path.

Question	Answer
How is score computed? Is risk just the score?	Frozen weighted formula; risk is computed independently from states; UNVERIFIABLE counts as 0 in the denominator.
Can bidders be ranked by score?	No. Score is a verification summary. Price evaluation is separate.
What if the tender changes?	Each audit record stores tender version and rule-set version; old cases are re-openable under the rules used.
Is the hash chain just blockchain?	No blockchain. It is a hash chain plus insert-only role, demonstrable by tampering with a row.
Real-world accuracy?	Tested only on synthetic data; we report counts observed, not real-world accuracy.
Who maintains the rules?	The organisation's procurement/compliance team maintains the versioned library; tender applicability is configured per tender.